

Lecture 2: Multinomial Distribution and ANOVA

Multinomial Distribution

Let $\underline{X} \sim \text{Multinomial}_k(n, \underline{p})$.

The MLE is $\hat{\underline{p}} = \frac{1}{n}\underline{X}$.

- $E[\hat{p}_i] = p_i$
- $X_i \sim \text{Bin}(n, p_i)$
- $\hat{p}_i = \frac{X_i}{n}$
- $\text{Var}(\hat{p}_i) = \frac{p_i(1-p_i)}{n}$

Covariance of $\hat{p}_i$ and $\hat{p}_j$

We are looking for $\text{Cov}(\hat{p}_i, \hat{p}_j)$. This is a 2-dim marginal problem. The joint PMF for (X_i, X_j) is:

$$P(X_i = x_i, X_j = x_j) = \frac{n!}{x_i!x_j!(n-x_i-x_j)!} p_i^{x_i} p_j^{x_j} (1-p_i-p_j)^{n-x_i-x_j}$$

We know $\text{Cov}(X_i, X_j) = E[X_i X_j] - E[X_i]E[X_j]$.

$$\begin{aligned} E[X_i X_j] &= \sum_{x_i, x_j} x_i x_j \frac{n!}{x_i!x_j!(n-x_i-x_j)!} p_i^{x_i} p_j^{x_j} (1-p_i-p_j)^{n-x_i-x_j} \\ &= \sum_{x_i \geq 1, x_j \geq 1} \frac{n(n-1)(n-2)!}{(x_i-1)!(x_j-1)!(n-x_i-x_j)!} p_i p_j p_i^{x_i-1} p_j^{x_j-1} (1-p_i-p_j)^{n-x_i-x_j} \\ &= n(n-1)p_i p_j \sum \frac{(n-2)!}{(x_i-1)!(x_j-1)!(n-x_i-x_j)!} p_i^{x_i-1} p_j^{x_j-1} (1-p_i-p_j)^{n-x_i-x_j} \\ &= n(n-1)p_i p_j \cdot 1 \\ &= n(n-1)p_i p_j \end{aligned}$$

The sum is 1 because it's the PMF of a Multinomial($n-2, (p_i, p_j, 1-p_i-p_j)$).

Now, we can find the covariance:

$$\begin{aligned}\text{Cov}(X_i, X_j) &= E[X_i X_j] - E[X_i]E[X_j] \\ &= n(n-1)p_i p_j - (np_i)(np_j) \\ &= n^2 p_i p_j - np_i p_j - n^2 p_i p_j \\ &= -np_i p_j\end{aligned}$$

Finally, for the MLE:

$$\text{Cov}(\hat{p}_i, \hat{p}_j) = \text{Cov}\left(\frac{X_i}{n}, \frac{X_j}{n}\right) = \frac{1}{n^2} \text{Cov}(X_i, X_j) = \frac{1}{n^2}(-np_i p_j) = \frac{-p_i p_j}{n}$$

Asymptotic Normality (CLT)

- Univariate: $\frac{\sqrt{n}(\hat{p}_i - p_i)}{\sqrt{p_i(1-p_i)}} \Rightarrow N(0, 1)$

- Multivariate CLT:

$$\sqrt{n} \Sigma^{-\frac{1}{2}}(\hat{p} - p) \xrightarrow{d} N_{K-1}(0, I)$$

Conditional Distribution

The conditional distribution $P(X_1 = x_1 | X_2 = x_2) = \frac{P(X_1=x_1, X_2=x_2)}{P(X_2=x_2)}$. This turns out to be:

$$(X_1 | X_2 = x_2) \sim \text{Bin}\left(n - x_2, \frac{p_1}{1 - p_2}\right)$$

- $n - x_2$: change of n
- $\frac{p_1}{1-p_2}$: scales, change of p_1

This comes from $p_1 + p_2 + \dots + p_k = 1$, so $p_1 + p_3 + \dots + p_k = 1 - p_2$.

Regression and ANOVA

- **Regression**: Predictors & Response both continuous.
- **Multinomial**: Distribution to model categorical random variable (unordered).
- **ANOVA**: Categorical predictors & continuous response.

(Reference: CH-15 of Linear Models With R by Faraway,
CH-16 of Practical Regression and Anova using R by Faraway)

In ANOVA, predictors are called **factors**. The response is continuous. The x_i takes discrete values (unordered).

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_k x_{ik} + \epsilon_i \quad (k \text{ factors})$$

We shall first talk about **one factor** ($k = 1$). The factor X_i takes discrete values, $j = 1, \dots, J$ (J levels). Levels: Values taken by x_i .

- Example: $y_i = \text{Height}$ (cont. response), $X_i = \text{Race}$ (one factor, $J = 5$ levels).
- We have observations at these different levels.

Example: One Factor, Two Levels ($J = 2$)

Consider a drug treatment group with 200 patients.

- 100 patients given medicine (Treatment group)
- 100 remaining patients not given medicine (Control group)
- Control group is given a **placebo** (sugar pill) to control for psychological effect.
- **Blind experiments:** Patient doesn't know which group.
- **Double-blind:** Doctors also don't know which group.

Let $y_i = \text{Time to recovery}$. We model this with Simple Linear Regression:

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i$$

Where $x_i = \begin{cases} 1 & \text{if in Treatment group} \\ 0 & \text{if in Control group} \end{cases}$

We want to test if the drug has an effect:

- $H_0 : \beta_1 = 0$ (Time to recovery is the same for both groups)
- $H_a : \beta_1 \neq 0$ (Time to recovery is the not same for both groups)

We can test this in several equivalent ways:

1. Fit Linear Regression, estimate β_1 , and use a t -test:

$$\hat{t} = \frac{\hat{\beta}_1}{s_{\hat{\beta}_1}} \sim t_{n-2} \quad (\text{e.g., } t_{198})$$

2. Use an F -test from ANOVA (which is t_{stat}^2).
3. Use a two-sample mean t -test.

Testing $\beta_1 = 0$ in this setup (2 levels) is the **same as** testing $H_0 : \mu_1 = \mu_2$ using a two-sample t -test (assuming equal variances, $s_p^2 \rightarrow \text{pooled}$).

Equivalence of T-test and Regression

Two-Sample T-test:

- Group 1: $X_1, \dots, X_m \sim N(\mu_1, \sigma^2)$
- Group 2: $Y_1, \dots, Y_n \sim N(\mu_2, \sigma^2)$

- $H_0 : \mu_1 = \mu_2$ vs. $H_a : \mu_1 \neq \mu_2$
- Test statistic: $t = \frac{\bar{X} - \bar{Y}}{s_p \sqrt{\frac{1}{m} + \frac{1}{n}}} \sim t_{m+n-2}$

Regression Model: Let the Control group ($x_i = 0$) be $X_1, \dots, X_m$ and the Treatment group ($x_i = 1$) be $Y_1, \dots, Y_n$.

- $X_i = \beta_0 + \epsilon_i \implies E[X_i] = \beta_0 = \mu_1$
- $Y_j = \beta_0 + \beta_1 + \epsilon_j \implies E[Y_j] = \beta_0 + \beta_1 = \mu_2$

Testing $H_0 : \mu_1 = \mu_2$ is the same as testing $H_0 : \beta_0 = \beta_0 + \beta_1$, which is $H_0 : \beta_1 = 0$.

The OLS estimate for $\hat{\beta}_1$ is the difference in means: We have:

$$t \sim t_{m+n-2}$$

To show:

$$t = \frac{\bar{X} - \bar{Y}}{s_p \sqrt{\frac{1}{m} + \frac{1}{n}}} \quad \text{same as} \quad \frac{\hat{\beta}}{s_{\hat{\beta}}}$$

Model Setup

$$\begin{aligned} x_i &= \beta_0 + \varepsilon_i, & i &= 1, \dots, m \\ y_i &= \beta_0 + \beta_1 + \varepsilon_i, & i &= m+1, \dots, n \end{aligned}$$

$$\text{Predictor: } \begin{bmatrix} 0 \\ \vdots \\ 0 \\ 1 \\ \vdots \\ 1 \end{bmatrix} \quad \begin{aligned} &\} 0 \text{ vector is } m \times 1 \\ &\} 1 \text{ vector is } n \times 1 \end{aligned}$$

Estimator of β_1

$$\hat{\beta}_1 = \frac{\frac{\sum y_i}{m+n} - \frac{n}{m+n} \cdot \frac{\sum x_i + \sum y_i}{m+n}}{\frac{n}{m+n} - \left(\frac{n}{m+n}\right)^2}$$

Simplifying,

$$\hat{\beta}_1 = \bar{y} - \bar{x}$$

Deriving the OLS Estimator $\hat{\beta}_1$ for Two Groups

We consider a model with two groups:

$$y_i = \begin{cases} \beta_0 + \varepsilon_i, & i = 1, \dots, m, \quad (x_i = 0), \\ \beta_0 + \beta_1 + \varepsilon_i, & i = m + 1, \dots, m + n, \quad (x_i = 1). \end{cases}$$

The total sample size is

$$N = m + n.$$

The ordinary least squares estimator for the slope in simple linear regression is

$$\hat{\beta}_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}.$$

Computing $\bar{x}$

Since m observations have $x_i = 0$ and n observations have $x_i = 1$,

$$\bar{x} = \frac{n}{m + n}.$$

Denominator

$$\sum (x_i - \bar{x})^2 = m(0 - \bar{x})^2 + n(1 - \bar{x})^2.$$

Substituting $\bar{x} = \frac{n}{m+n}$ gives

$$\sum (x_i - \bar{x})^2 = m \left(\frac{n}{m+n} \right)^2 + n \left(\frac{m}{m+n} \right)^2 = \frac{mn}{m+n}.$$

Numerator

Split the sum by groups.

Group 1 ($x_i = 0$)

$$(0 - \bar{x})(y_i - \bar{y}) = -\bar{x}(y_i - \bar{y}),$$

so the contribution is

$$-\bar{x} \sum_{i=1}^m (y_i - \bar{y}) = -\bar{x}(m\bar{y}_0 - m\bar{y}),$$

where $\bar{y}_0$ is the mean of group 0.

Group 2 ($x_i = 1$)

$$(1 - \bar{x})(y_i - \bar{y}) = (1 - \bar{x})(y_i - \bar{y}),$$

so the contribution is

$$(1 - \bar{x}) \sum_{i=m+1}^{m+n} (y_i - \bar{y}) = (1 - \bar{x})(n\bar{y}_1 - n\bar{y}),$$

where $\bar{y}_1$ is the mean of group 1.

Putting both groups together and simplifying yields

$$\sum (x_i - \bar{x})(y_i - \bar{y}) = \frac{mn}{m+n} (\bar{y}_1 - \bar{y}_0).$$

Final Estimator

Putting numerator and denominator together:

$$\hat{\beta}_1 = \frac{\frac{mn}{m+n}(\bar{y}_1 - \bar{y}_0)}{\frac{mn}{m+n}} = \bar{y}_1 - \bar{y}_0.$$

$$\boxed{\hat{\beta}_1 = \bar{y}_{\text{treatment}} - \bar{y}_{\text{control}}}$$

Thus, the OLS estimator of the slope in this two-level model equals the *difference in sample means*.

Interpretation

$$\hat{\beta}_1 = \bar{Y} - \bar{X}$$

The pooled variance estimator from the t -test is:

$$s_p^2 = \frac{\sum (X_i - \bar{X})^2 + \sum (Y_i - \bar{Y})^2}{m + n - 2}$$

The variance estimator $\hat{\sigma}^2$ (or MSE) from regression is **exactly the same** as s_p^2 . (H.W.)

$$\hat{\sigma}^2 = s_p^2$$

The design matrix $\mathbf{X}$ and its $\mathbf{X}^T \mathbf{X}$ are:

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ \vdots & \vdots \\ 1 & 0 \\ 1 & 1 \\ \vdots & \vdots \\ 1 & 1 \end{pmatrix} \implies \mathbf{X}^T \mathbf{X} = \begin{pmatrix} m+n & n \\ n & n \end{pmatrix}$$

The inverse is:

$$(\mathbf{X}^T \mathbf{X})^{-1} = \frac{1}{mn} \begin{pmatrix} n & -n \\ -n & m+n \end{pmatrix}$$

The variance of $\hat{\beta}_1$ is $\text{Var}(\hat{\beta}_1) = \sigma^2 (\mathbf{X}^T \mathbf{X})_{2,2}^{-1}$.

$$s_{\hat{\beta}_1}^2 = \hat{\sigma}^2 \cdot (\mathbf{X}^T \mathbf{X})_{2,2}^{-1} = s_p^2 \left(\frac{m+n}{mn} \right) = s_p^2 \left(\frac{1}{m} + \frac{1}{n} \right)$$

Therefore, the t -statistic for $\hat{\beta}_1$ is:

$$t = \frac{\hat{\beta}_1}{s_{\hat{\beta}_1}} = \frac{\bar{Y} - \bar{X}}{s_p \sqrt{\frac{1}{m} + \frac{1}{n}}}$$

This is identical to the two-sample t -statistic.